HARSHIT KUMAR

Major - Data Science
Roll no : IMS22280
Indian Institute of Science Education and Research
Thiruvananthapuram

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Education

- **Indian Institute of Science Education and Research Thiruvananthapuram** *(2022-2027)*
  Integrated BS-MS Kerala
  - Major(Data Science) - 7.5 CGPA

  IISER Thiruvananthapuram is a premier Tier-1 autonomous institute of national importance, recognized for cutting-edge scientific research, rigorous interdisciplinary training, and its strong standing in India's top NIRF-ranked institutions.

Experience & Projects

- **Spike Agreement–Dependent Plasticity (SADP): Unsupervised**
  Researcher / Co-Author (School of Data Science) June 2025 – Feb 2026
  - Proposed a linear-time learning rule replacing STDP (**Published in Neural Networks**).

- **Spike Agreement–Dependent Plasticity (SADP): Supervised**
  Researcher / Co-Author (School of Data Science) Feb 2026 – Ongoing
  - Built hybrid CNN–SNN SADP (**Preprint arXiv:2601.08526v2**, under review in Neural Networks).

- **NewsBits Software** *Thiruvananthapuram, Kerala*
  Automated Document Segmentation Software for Kerala Government Summer 2025
  - Built newspaper article segmentation software for Kerala Government (**Sample Images**).

- **GLARE: Graph Learning Through Affinity-Guided Rewiring**
  Research Project (Submitted to a Conference) Ongoing
  - Developed affinity-guided rewiring for heterophilic graphs (**Preprint arXiv:2609.32613**).

- **HyperFuse: Fast Self-Supervised Node Embeddings**
  Research Project (Submitted to Machine Learning, Springer) Submitted
  - Proposed label-free hypergraph node embeddings, up to 179x faster than baselines.

- **EUREKA: Label-Free Edge-Utility Rewiring**
  Research Project (Submitted to Data Mining and Knowledge Discovery) Submitted
  - Built label-free edge-utility rewiring for heterophilic graphs, state-of-the-art gains.

- **DELTA: Multi-Domain Adaptation for Medical Anomaly Detection**
  Research Project (Preparing for Submission) Ongoing
  - Developing multi-domain adaptation for medical anomaly detection on unseen domains.

- **Food-Web Structure Anticipates Ecosystem Recovery After Collapse**
  Researcher (School of Data Science) Ongoing
  - Built a GNN to predict species recovery from food webs.

- **Image-to-Songs Recommendation Model: Conference Paper**
  Multimodal Deep Learning Research Project IEEE ICRM 2025 (Published & Presented)
  - Built image-to-song recommender with ResNet50, BLIP, Sentence-BERT (**Paper** | **Report**).

- **Summer Internship - IIT Patna** *Patna, Bihar*
  Pharmacovigilance Using LLMs May 2024 – July 2024
  - Analyzed drug side-effects; contributed to a pharmacovigilance LLM (**Certificate**).

Technical Skills

- **Programming Languages and Tools**: Python, C, C++ (familiar), R (familiar), Keras, PyTorch, Pandas, NumPy, Scikit-learn, Tesseract, Tableau
- **Machine Learning**: Random Forest, SVM, Boosting (GBM, AdaBoost), Clustering (K-Means), Dimensionality Reduction (PCA, t-SNE, UMAP), Optimization Techniques
- **Deep Learning**: CNN, LSTM, GRU, Autoencoders, GANs, Transformers, Large Language Models (LLMs), Graph Neural Networks (GNN, GCN, GAT), U-Net, ResNet, EfficientNet
- **Data Structures & Algorithms**: Graphs (BFS, DFS, Dijkstra), DP, Greedy, Backtracking, Union-Find
- **Other Courses**: AI and Medical Imaging, Big Data Analytics, Introduction to Large Language Models (LLMs), Parallel and Distributed Computing, Statistical Modelling, Survival Analysis, Bioinformatics, Systems Biology

Achievements

- **Scholarships:** Awarded the BVC Merit-Based Scholarship, which covers 50% of the semester tuition fee based on academic performance.
- **Conference Publication:** Research paper selected, published, and presented at the IEEE International Conference on Robotics and Mechatronics (ICRM) 2025.
- **Research Publication:** SADP-based neuromorphic learning research selected for publication at *Neural Networks* and another related work currently under review at *Neurocomputing*.

Skills

- **Digital Skills**: Tableau, Digital Art, Canva, LaTeX, PowerPoint, Linux, MS Excel, Google Sheets
- **Management Skills:** Active member of the SCoM (Student Committee of Mess) at IISER TVM, contributing to coordination and decision-making for mess operations.
- **Event Leadership:** Led the Design and Decoration Team for Diwali 2025 celebrations at IISER TVM, managing planning, coordination, and execution.
- **Project Coordination:** Experience managing and organizing multiple Machine Learning and Tableau projects as part of academic coursework and collaborative research.